Product Teardown: DocAssist

Device: Web app (Python, Flask), clinical setting

Goal: Cut clinical note reading from 15 minutes to under 10 seconds

Read: 4 minutes

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DocAssist's Overview

DocAssist

DocAssist is a **clinical documentation assistant**. It turns long clinical notes and medical documents into short summaries and answers a doctor's follow-up questions, using open-source LLMs served through the Groq API.

Python · Flask · Groq API · Mistral · NLP pipeline · Rubric evaluation

Key Insights:

Speed: manual reading of about 15 minutes becomes a summary in under 10 seconds.

Validated: tested with 15+ real hospital users before it was called done.

Graded, not trusted: every response is scored against a rubric for relevance and safety.

The problem

DocAssist

Pain point	Reality
Symptom confusion	Patients cannot tell if symptoms need urgent care.
Manual documentation	Doctors spend hours on paper-based clinical workflows.
Zero continuity	Each hospital interaction starts from scratch.
No vital tracking	Vitals are tracked on paper, separate for each visit.

Who it is for

D

Doctor

Primary reader

Faces long clinical notes and reports. Needs a **short, trustworthy summary** and quick answers to follow-up questions, with the **patient history in one place.**

P

Patient

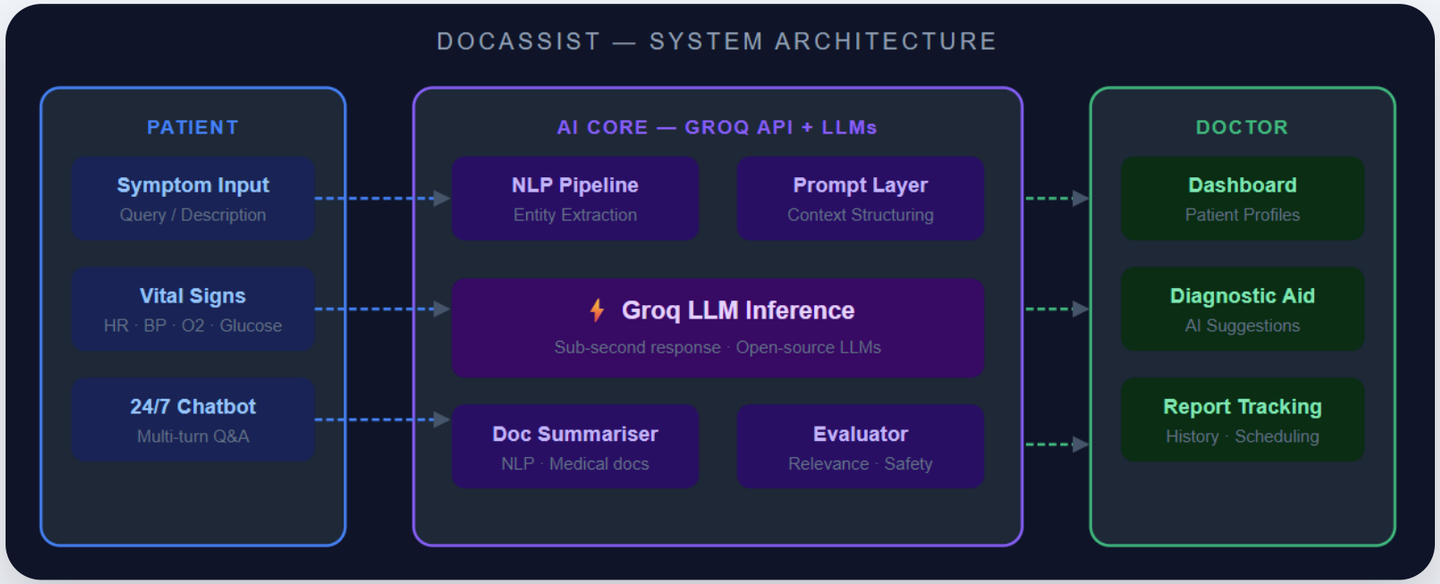
Input side

Describes symptoms and shares vitals. Needs a **24/7 assistant** that remembers the conversation and helps decide whether to seek care.

System architecture

DocAssist

Three layers: the patient layer captures symptoms, vitals and queries; the AI core runs entity extraction, prompt structuring, Groq inference, summarisation and evaluation; the doctor layer shows a dashboard, diagnostic aid and report tracking.



Patient flow

DocAssist

01
Patient input
Symptoms or a query.

→

02
NLP pipeline
Entity extraction.

→

03
Groq LLM
Fast inference.

→

04
Evaluation
Relevance and safety.

→

05
Doctor review
AI-assisted summary.

The README describes the journey as five automated steps, from symptom input to a doctor-ready summary in under 60 seconds.

Key features

1

24/7 chatbot

Symptom assessment with multi-turn memory.

2

Document summaries

An NLP pipeline turns medical documents into plain-language summaries.

3

Doctor tools

Diagnostic suggestions, vital signs dashboard, patient profiles and report tracking.

4

Evaluation engine

Rubric-based scoring of each response, with a feedback loop.

Why Groq: latency

DocAssist

~3.2s

per turn on GPT-4

README benchmark

~0.4s

per turn on Groq LLaMA 3

README benchmark

15min to 10s

document summarisation

Standard inference added 2 to 4 seconds per turn, too slow for live clinical use. Sub-second inference made real-time use viable.

Results

DocAssist

Metric	Baseline	DocAssist	Note
Response relevance	~60%	~90%	+30%, across 100+ interactions (README)
Document summarisation	Manual, 15 min	Under 10 s	Fully automated
Hospital validation	None	15+ real users	Tested in a live clinical setting
Competition		Top 25 pan-India	Unleash LLM Challenge

What it became

>

Rubric and failure taxonomy

Grading LLM outputs for hallucination, reasoning and output quality produced a reusable rubric.

>

Judge agent

That rubric and taxonomy later became the judge agent in the CupShup harness.

>

Evaluation habit

At 1M1B, 100+ LLM outputs were graded with LLM-as-a-Judge.

Success metrics

DocAssist

NORTH STAR METRIC

Clinician time saved per note, with no drop in rubric score

Proposed. The 15 minutes to under 10 seconds claim is the headline; this keeps quality attached to it.

L1 METRIC (PROPOSED)

Share of summaries a doctor accepts without edits.

L2 METRIC (PROPOSED)

Rubric score and response latency per turn, tracked on every call.

Thank you for reading.

README: github.com/godhulivyas-build/DocAssist

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DISCLAIMER

DocAssist has no public live UI, so this deck uses the architecture diagram and the README, not screenshots. Success metrics are proposals, not measured results.

DocAssist